Methodology

The analysis began with comprehensive data preprocessing. Column names were cleaned and standardized to ensure consistency across both datasets. All timestamps were converted to a daily level so that trading activity could be accurately aligned with sentiment data. The trading dataset was then merged with the corresponding daily sentiment classification from the Fear & Greed Index.

After merging, key daily performance metrics were engineered to capture trader behavior more effectively. These included daily profit and loss (PnL), trade frequency, win rate, and average position size. A sentiment-based comparative analysis was conducted to evaluate how these metrics varied across different market regimes. Visualizations were developed to highlight behavioral trends and performance differences under Fear, Neutral, and Greed conditions.

Key Insights

The results reveal clear regime-dependent patterns in trader behavior. Trade frequency increases significantly during Greed and Extreme Greed periods, suggesting higher participation and confidence when market sentiment is optimistic. In contrast, PnL volatility rises during Fear regimes, reflecting increased uncertainty and heightened risk exposure.

The analysis also indicates that overtrading during high-volatility periods tends to reduce consistency in performance. Overall, the findings demonstrate that market sentiment has a measurable influence on short-term trader performance and risk-taking behavior.

Strategy Recommendations

Based on these insights, a sentiment-aware risk management framework is recommended. During Extreme Fear conditions, leverage and exposure should be reduced to limit downside risk. Conversely, exposure may be scaled selectively during Greed regimes when market conditions appear favorable.

Additionally, implementing trade frequency controls during volatile regimes can prevent excessive risk-taking. Finally, incorporating regime-based capital allocation overlays can improve risk-adjusted returns by dynamically adjusting portfolio exposure according to prevailing market sentiment.